key components, and recent developments. I then guide you through getting started with React and Next.js for AI applications, followed by a deep dive into the Vercel AI SDK for handling streaming responses, working with multiple AI providers, and managing state with React server components.

- Part 2: Advanced generative AI techniques and deployment (chapters 5–9) moves into more sophisticated methods and crucial deployment considerations. You'll master prompt engineering techniques; learn to use LangChain.js for complex AI workflows and agents; and understand the intricacies of testing, debugging, and securing your generative AI applications for production.
- Part 3: Hands-on projects (chapters 10–11) puts everything into perspective through comprehensive project explorations. I provide full codebases for an AI interview assistant and an AI RAG agent and then dissect their features, architectural decisions, and critical development considerations.
- Part 4: Advanced integrations and the future of AI (chapter 12) moves beyond current best practices to look at what's next. This part is dedicated to the Model Context Protocol, an emerging standard designed to create more seamless, secure, and interoperable connections between your applications and various AI models.

While the book is structured for a linear read to build knowledge sequentially, experienced developers might choose to focus on specific chapters within parts 2 and 3 that align with their immediate interests after completing part 1. The projects in part 3 serve as excellent consolidation exercises and practical references.

About the code

This book contains many examples of source code both in numbered listings and in line with normal text. In both cases, source code is formatted in a `fixed-width font like this` to separate it from ordinary text. Sometimes code is also **`in bold`** to highlight code that has changed from previous steps in the chapter, such as when a new feature adds to an existing line of code.

In many cases, the original source code has been reformatted; we've added line breaks and reworked indentation to accommodate the available page space in the book. In some cases, even this was not enough, and listings include line-continuation markers (➥). Additionally, comments in the source code have often been removed from the listings when the code is described in the text. Code annotations accompany many of the listings, highlighting important concepts.

You can get executable snippets of code from the liveBook (online) version of this book at https://livebook.manning.com/book/build-ai-enhanced-web-apps. The complete code for the examples in the book is available for download from the Manning website at https://www.manning.com/books/build-ai-enhanced-web-apps, and from GitHub at https://github.com/Generative-AI-Web-Apps/Code.

Full details for running the examples are in the appendix. To run the code examples and projects in this book, you will need